

A Low-Power Integrated Bioamplifier With Active Low-Frequency Suppression

Benoit Gosselin, *Student Member, IEEE*, Mohamad Sawan, *Fellow, IEEE*, and C. Andrew Chapman

Abstract—We present in this paper a low-power bioamplifier suitable for massive integration in dense multichannel recording devices. This bioamplifier achieves reduced-size compared to previous designs by means of *active low-frequency suppression*. An active integrator located in the feedback path of a low-noise amplifier is employed for placing a highpass cutoff frequency within the transfer function. A very long integrating time constant is achieved using a small integrated capacitor and a MOS-bipolar equivalent resistor. This configuration rejects *unwanted low-frequency contents* without the need for input RC networks or large feedback capacitors. Therefore, the bioamplifier high-input impedance and small size are preserved. The bioamplifier, implemented in a 0.18- μm CMOS process, has been designed for neural recording of action potentials, and optimised through a transconductance-efficiency design methodology for micropower operation. Measured performance and results obtained from *in vivo* recordings are presented. The integrated bioamplifier provides a midband gain of 50 dB, and achieves an input-referred noise of 5.6 μVrms . It occupies less than 0.050 mm² of chip area and dissipates 8.6 μW .

Index Terms—Active low-frequency suppression, biopotential recording, dense integrated neural interfaces, integrated bioamplifier, micropower analog circuit design.

I. INTRODUCTION

MULTISITE micromachined probes have been coupled with advanced microsystems to build implantable devices capable of tackling the central nervous system at the cellular level [1]–[5]. Because such sensors enable *in situ* recordings of the simultaneous bioelectrical activity from multiple individual cortical neurons for long periods of time, they are becoming powerful tools for various areas within neuroscience research [6], [7]. These developments have been stimulated by astonishing emerging clinical applications including mind-driven rehabilitation systems and implants for treating diseases such as epilepsy [8], [9].

But, the number of recording sites provided by micromachined probes is always growing, and sensors that incorporate high channel-count are needed to accommodate this expansion. The required interfaces must include data acquisition circuits comprising low-noise amplifiers, filters, data converters and

processing circuitry, such as biopotential detector, for performing *in vivo* data reduction [5]. Data reduction is intended to decrease the data rate that must be handled by low-power telemetry for enabling wireless operation of implanted sensors. Nonetheless, integrated circuits that interface with micromachined probes are subject to stringent size and power restrictions due to the probe shape and vulnerability of the tissue. Circuits must ideally fit within silicon areas as low as 0.160 mm² per channel because a 400- μm inter-electrode spacing is commonly used for neural recording arrays. In addition, the power dissipation must be restrained below a few hundred microwatts per channel to avoid harming cortical tissue [10].

Although bioamplifiers that retrieve weak bioelectrical signals have been developed extensively [1]–[3], [5], [11]–[15], very few reported designs meet the noise, power, and size requirements for massive integration in implantable multichannel recording devices. Furthermore, the integrated designs that have been proposed entirely fill or exceed the available area, and dissipate too much power for safe permanent usage. In several cases, circuitry needed to remove low-frequency components such as the dc offset across differential sensing electrodes accounts for much of the total area of the circuit. This can restrict the number of recording channels that can be handled by the sensor and/or prevent introduction of other on-chip modules.

Electrode potential mismatch is responsible for input offsets that can reach hundreds of millivolts [12]. Among *low-frequency suppression* schemes, only few configurations were considered for on-chip integration in implantable multichannel sensors. First, it has been shown that approaches using a capacitor feedback network [2], [3], [5], [12], [14] can degrade the input-referred noise, lower the input impedance of the amplifier, and can be subject to mismatch in passive input devices. More significantly, this configuration requires very large integrated capacitors that dominate the amplifier area in order to achieve satisfactory gains. A second approach loads the recording site with a high-value resistor, implemented either with a reverse-biased diode [1], [11], or with subthreshold transistors [15], to form a highpass filter with the electrode capacitance. This technique is affected by the electrical parameters of electrodes, which vary from one site to another, and this alters cutoff frequencies. Moreover, this scheme requires additional biasing circuitry and does not provide common mode rejection unless other circuit components are added. Both schemes make use of ac coupling of input devices that are subject to mismatch, and this can degrade input impedance and may lead to signal attenuation. Active schemes based on closed-loop control of dc levels have been proposed and tested with commercial off-the-shelf integrated circuits [16], [17].

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B. Gosselin and M. Sawan are with the Polystim Neurotechnologies Laboratory, Electrical Engineering Department, École Polytechnique de Montréal, Montréal, QC H3C 3A7, Canada (e-mail: benoit.gosselin@polymtl.ca).

C. A. Chapman is with the Center for Studies in Behavioral Neurobiology, Department of Psychology, Concordia University, Montréal, QC H4B 1R6, Canada.

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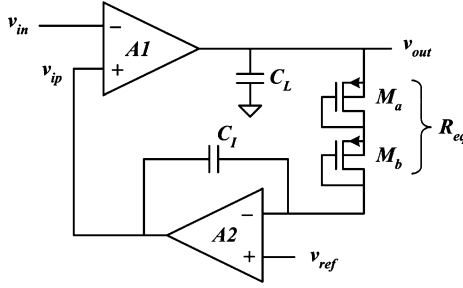


Fig. 1. Schematic of the bioamplifier with its low-frequency suppression circuit made of a Miller integrator connected between the amplifier output (v_{out}) and positive input (v_{ip}). The bioamplifier uses negative feedback since a negative sign is attached to the inverting integrator output voltage.

Such active schemes enhance *low-frequency suppression* and allow for high input impedance.

We present in this paper the design and performance of a new low-power integrated bioamplifier that implements active low-frequency suppression. The bioamplifier features a bandpass characteristic and avoids passive ac couplings input networks. The implemented design fits within the smallest reported area, and has one of the best noise versus power consumption tradeoffs.

II. ARCHITECTURE OF THE BIOAMPLIFIER

Fig. 1 shows a schematic of the proposed bioamplifier with its active *low-frequency suppression* circuit. First introduced in [18], this design consists of a low-noise amplifier (A1) featuring an inverting Miller integrator within its feedback path. The active Miller integrator is made from a second amplifier (A2), a capacitor (C_I) and a high-value resistor (R_{eq}). Its time constant (τ) sets the -3 -dB highpass cutoff frequency (f_{hp}) of the bioamplifier ($\tau = R_{eq}C_I$). The midband gain of the bioamplifier equals the low-frequency open-loop gain of A1, noted A_{01} , and the -3 -dB lowpass cutoff frequency (f_{lp}) is defined by A1 dominant pole. Despite the fact that the feedback loop is connected from A1 output to its positive input terminal (v_{ip}), the bioamplifier uses negative feedback since the inverting integrator attaches a negative sign to its output voltage. A2 senses the reference voltage at node v_{ref} and replicates it at v_{ip} for providing common mode voltage rejection in the bioamplifier. Here, voltage-to-voltage amplifier topologies are used for implementing A1 and A2.

A. Low-Frequency Suppression Circuit

The Miller integrator samples the main amplifier (A1) output voltage at node v_{out} , integrates it, and applies a correction voltage at terminal v_{ip} . As a result, node v_{ip} tracks the dc level seen by A1 at v_{in} , which effectively cancels out any dc offset, drift and *other low-frequency input voltages*. As depicted in Fig. 2, the active integrator yields a loop gain greater than unity from node v_{out} to v_{ip} at low frequencies, but provides a strong attenuation in the signal bandwidth. Its magnitude decreases with 20 dB/decade and its phase is 90° . An ideal integrator would present an infinite gain at dc, but the practical integrator exhibits a dc gain that equals the dc gain of amplifier A2 (noted A_{02}). The integrator corner frequency is at $\omega = 1/(\tau A_{02})$. The frequency response shape of A2, also reported in Fig. 2, shows the open-loop bandwidth (ω_a) and

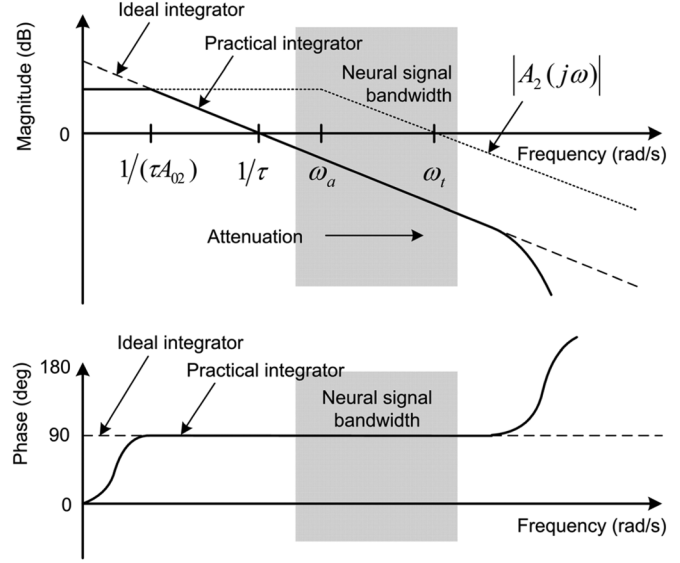


Fig. 2. Magnitude and phase responses of the Miller integrator. Output signal undergoes strong attenuation for frequencies beyond $1/\tau$, and a 90 degrees phase shift is introduced. Ideal and practical integrators cases are shown for comparison.

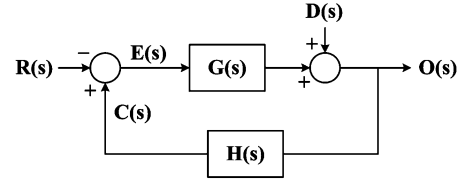


Fig. 3. Equivalent feedback control system for the bioamplifier transfer function, with input $R(s)$, error $E(s)$, control signal $C(s)$, and disturbance $D(s)$.

the unity-gain frequency (ω_t) of amplifier A2. Note that A_{02} must be made as large as possible to attenuate non-ideal effects such as excess phase in the integrator.

As shown in Fig. 1, a MOS-bipolar device, which consists of transistors M_a and M_b , implements R_{eq} . Equivalent resistors greater than $10^{13} \Omega$ are achieved for small input signals ($|\Delta V| < 100$ mV) with this configuration [19]. A negative gate to source voltage ($V_{GS} < 0$), makes M_a and M_b to act as diode connected pMOS transistors, whereas a positive V_{GS} turns MOS transistors off, activates the parasitic source-well-drain p-n-p bipolar junction transistors, and changes M_a and M_b into diode connected BJTs. Both operating modes yield diodes like exponential I - V relationships, presenting high resistive characteristics near the origin. Therefore, very long time constants, requiring only small-integrated capacitor sizes, can be implemented. A metal-to-metal (MiM) capacitor of 1 pF is used for this design, and M_a and M_b are sized to $4 \mu\text{m} \times 4 \mu\text{m}$.

B. Frequency Response of the Bioamplifier

Fig. 3 shows a functional block diagram of the bioamplifier transfer function. Ideal amplifiers with no disturbance ($D(s) = 0$) are assumed, and $G(s) = A_{01}$, $H(s) = -1/(\tau s)$. Then, the transfer function to the ideal bioamplifier is

$$\frac{O(s)}{R(s)} = \frac{-G(s)}{1 - G(s)H(s)} = \frac{-s\tau A_{01}}{s\tau + A_{01}}. \quad (1)$$

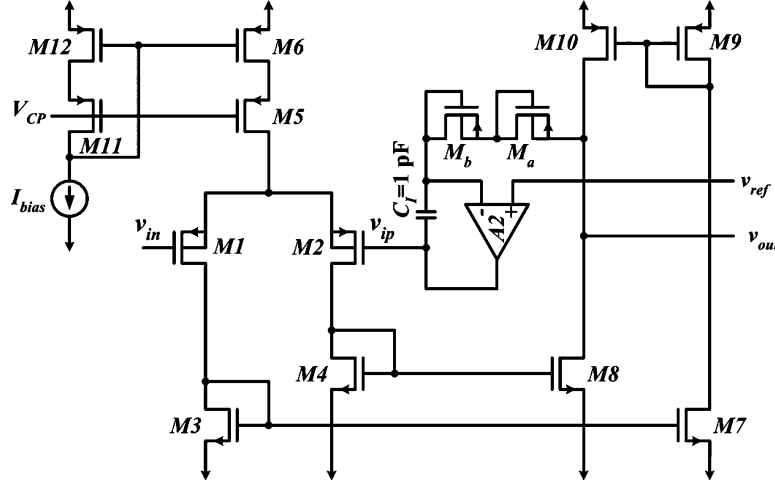


Fig. 4. Schematic of the current mirror operational transconductance amplifier A1 and the Miller integrator-based low-frequency suppression circuit.

This transfer function yields a -3 dB highpass cutoff frequency, given by

$$f_{hp} = \frac{1}{2\pi} \frac{A_{01}}{R_{eq}C_I}. \quad (2)$$

Equation (2) shows that f_{hp} is shifted upward on the frequency axis according to the gain of amplifier A1. This is particularly helpful in the targeted application since the useful bandwidth of neural signals only starts at a few hertz.

Besides, the presented circuit can be regarded as a modified proportional-integral-derivative (PID) controller that implements integral control action. Letting $s \rightarrow 0$ in (1) and in $O(s)/D(s)$ (Fig. 3) shows that the suggested scheme eliminates low-frequency input signals and disturbances as both expressions tend to 0. Thus, in addition to remove low input frequencies and offset that results from electrodes mismatch, this scheme also cancels any internal offsets due to integrated devices mismatch and process variations in amplifiers A1 and A2. Moreover, drift from the electrodes and from the amplifiers are also attenuated.

III. BIOAMPLIFIER CIRCUIT DESIGN

The bioamplifier circuit schematic is shown in Fig. 4. It is designed for recording action potentials (APs) and uses two operational transconductance amplifiers (OTAs). Extracellular APs are weak amplitude signals (50 – 500 μ V) in the 100 Hz – 10 kHz frequency band recorded in the vicinity of a neuron. A current mirror OTA was chosen as the main amplifier A1 because it relies only on a few transistors, thus making it a good candidate for low-noise and low-power performances and compact implementation. A1 low-frequency gain is $A_{01} = g_m/g_o$, and dominant pole is $\omega_{p1} = g_m/(A_{01}C_L)$, where output conductance g_o results from output transistors $M_{8,10}$ ($g_o = g_{ds8} + g_{ds10}$), and OTA transconductance is $g_m = g_{m1,2}$. The load capacitor C_L is set to 400 fF, putting f_{hp} around 9 kHz for this design. Unity-gain current mirrors are used, i.e., $I_{D1,2} = I_{D3,4} = I_{D7,8} = I_{D9,10}$.

In contrast, a two-stage OTA with lead-compensation is used to implement A2 in the feedback path. This topology was chosen because it provides a high dc gain in the Miller integrator, and only requires a small capacitor and a resistor for compensation

[20]. Moreover, it is very compact, requiring fewer transistors than the current mirror OTA used for A1. A lead-compensation network made of a 190 k Ω resistor and a 2 pF capacitor is used for this OTA. The high dc gain of A2 results from the cascade of a differential pair gain stage and a common source gain stage. The gain of each stage is obtained by dividing the transconductance of the stage by the small signal conductance of its output load [20]. Multiplying the gains of both stages together gives A_{02} , which is set to around 20 kV/V in this design.

A. Micropower Amplifier Design Methodology

Designing for low-noise performances under the aforementioned power restrictions is a challenging task. Hence, amplifiers were optimised through a transconductance efficiency-based design methodology. This design strategy allows for optimal sizing of transistors for operation over the entire continuum of channel inversion levels, with respect to a pre-selected drain current and inversion coefficient (IC). The later results from the normalized drain current I_D divided by the specific current I_S and is proportional to the level of channel inversion of a MOS transistor [21]

$$IC = \frac{I_D}{2n\mu C_{ox} \left(\frac{W}{L}\right) U_T} \quad (3)$$

where n is the slope factor, μ is the carrier mobility, C_{ox} is the gate-oxide capacitance per unit area, W and L are the channel width and length, and U_T is the thermal voltage (kT/q). The MOS transconductance efficiency (g_m/I_D) is maximum for transistors operating in weak inversion ($IC < 0.1$) and typically reaches its maximum at 25.5 V $^{-1}$, in bulk CMOS processes, for very low IC. Then, going from weak inversion to moderate inversion ($0.1 < IC < 10$), g_m/I_D monotonically decreases, and reaches its minimum in strong inversion ($IC > 10$). Specifically, operation at very low IC results in optimally high transconductance for a given current, but requires large W/L ratios. Conversely, operating at high IC yields low transconductances, but offers larger bandwidth. We use expression

$$\frac{g_m}{I_D} \cong \frac{1}{nU_T} \varphi(IC) \quad (4)$$

valid for all levels of inversion, where

$$\varphi(IC) = \frac{1}{0.5 + \sqrt{0.25 + IC}} \quad (5)$$

is taken from the EKV model [22], and yields values between 0 and 1. The design method first consists of choosing transistors drain currents according to power and noise specifications. Then, the amplifier parameters such as input-referred noise, low-frequency gain, output resistance, and bandwidth are optimized through careful selection of devices size according to the required channel inversion level.

B. Low-Noise OTA Optimization

The input-referred thermal noise power ($v_{ni,thermal}^2$) achieved by OTAs (A1 and A2) are proportional to transconductance ratios of transistors in the differential pairs and in the current mirrors. For long-channel devices, we can show that $v_{ni,thermal}^2$ for OTA A1 is

$$v_{ni,thermal(A1)}^2 = 2v_{n1}^2 + 4v_{n3}^2 \left(\frac{g_{m3}}{g_{m1}} \right)^2 + 2v_{n9}^2 \left(\frac{g_{m9}}{g_{m1}} \right)^2 \quad (6)$$

where v_{ni}^2 equals $4kTn\Gamma_i/g_{mi}\Delta f$, and the thermal noise factors Γ_i take values 1/2 or 2/3, whether the device operates in weak or strong inversion. Then, assuming n constant for all regimes of operation, and replacing v_{ni}^2 and (5) in (6) gives

$$v_{ni,thermal(A1)}^2 = \left[8kT \frac{n^2 U_T}{I_{D1}} \frac{1}{\varphi(IC_1)} \cdot \left(\Gamma_1 + 2 \frac{\varphi(IC_3)}{\varphi(IC_1)} \Gamma_3 + \frac{\varphi(IC_9)}{\varphi(IC_1)} \Gamma_9 \right) \right] \Delta f_{(A1)}. \quad (7)$$

Equation (7) shows that, for minimizing the noise, high IC are required in the differential pair (linked to $\varphi(IC_1)$) whereas low IC are needed for the current mirrors/active loads (linked to $\varphi(IC_{3,9})$). Operating $M_{1,2}$ in weak inversion (i.e., pushing $\varphi(IC_1)$ toward 1), and the remaining transistors in strong inversion (i.e., pushing $\varphi(IC_3)$ and $\varphi(IC_9)$ downward 0) fulfills this requirement. This is achieved by picking large $(W/L)_{1,2}$ and low $(W/L)_{3,4,7,8,9,10}$ ratios, with respect to a fixed drain current.

Similar design considerations apply for A2, but in this case, the noise power only depends on two IC (related to the differential pair and the active load, respectively). Under these design considerations, A1 and A2 input thermal noises simplify to

$$v_{ni,thermal(OTA)}^2 = 4kT \frac{n^2 U_T}{I_{D1}} \frac{1}{\varphi(IC_1)} \Delta f. \quad (8)$$

Equation (8) is used as a design insight to set the transistors drain current according to satisfactory noise performance. Bias currents (I_{bias}) of 1 μA and 2 μA are used in the cascode current sources for biasing A1 and A2, respectively (Fig. 4). This yields an input noise voltage of around 3 μV_{rms} for A1 at body temperature, according to (8). A higher bias current is used for A2, ensuring stability for closed-loop operation.

Besides, large area pMOS transistors with small inversion coefficients are used in the differential input pairs of A1 and A2 for minimizing the $1/f$ noise in the bioamplifier. In general,

TABLE I
DEVICES GEOMETRY AND OPERATING POINTS FOR OTA A1

Devices	W/L (μm)	I_D (μA)	IC	g_m/I_D (V^{-1})	V_{eff} (V)
M_1, M_2	800.00/4.00	0.5	0.035	24.9	-0.123
$M_3, M_4,$ M_7, M_8	2.00/20.00	0.5	17.1	6.37	0.277
$M_5, M_6,$ M_{11}, M_{12}	1.50/2.16	1.0	3.86	11.8	0.122
M_9, M_{10}	2.00/20.00	0.5	80.3	3.13	0.604

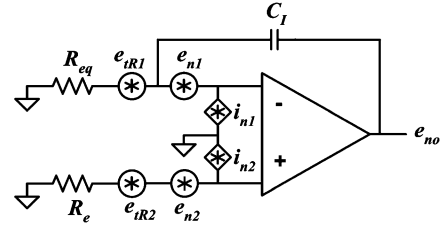


Fig. 5. Equivalent model for determining voltage gains of noise sources in the Miller integrator.

pMOS transistors exhibit less $1/f$ noise than nMOS devices as long as their gate voltage overdrive ($V_{eff} = V_{GS} - V_{TH}$) stays low. Even though this effect appears to be more substantial in larger geometry processes [23], the $1/f$ noise level at low V_{eff} is approximately half an order of magnitude less in pMOS devices than in nMOS, for 0.18- μm bulk CMOS technology [24]. Simulated corner frequencies of a few hundred hertz are achieved for both OTAs using large input transistors ($L_{1,2} = 800 \mu m$, $W_{1,2} = 4 \mu m$), with BSIM3 version 3.2 level 11 SPICE transistor model and *Spectre* simulator. Having such low corner frequencies reduces the impact of the $1/f$ noise and makes the overall input noise to mainly depend on the transistors thermal noise [25].

Table I shows the dc operating points and dimensions of transistors used in A1. Note that cascode transistors M_5 and M_6 are operated in moderate inversion for achieving good matching of current mirrors and high output impedances in the current sources. Such operation conditions increases the common mode rejection ratio (CMRR) of A1 [20]. Also, note that long-channel transistors (above 2 μm) are used for all devices in this design in order to avoid excess thermal noise in transistors due to channel-length modulation and small-geometry effects such as carrier heating and velocity saturation [26].

C. Total Input-Referred Noise

The proposed bioamplifier allows direct coupling of neural signals to its input terminals and therefore involves no resistive or capacitive voltage divider or other input coupling network. This prevents the input signal from attenuation and avoids degradation of the signal-to-noise ratio. In fact, A1 and the *Miller integrator* both contribute to the total input-referred noise of the bioamplifier. An expression for the noise contribution of the integrator can be extracted from the noise model shown in Fig. 5.

Thermal noise sources are associated to resistor R_{eq} , and the resistive part of the reference electrode (R_e) connected to terminal v_{ref} (Fig. 1), denoted e_{tR1} and e_{tR2} , respectively, on Fig. 5. The OTA also comprises intrinsic noises sources (e_n and

i_n), reported on Fig. 5 as well. The noise voltage density e_n is composed of $A2$ input thermal and $1/f$ noises (the later contribution is neglected assuming a low corner frequency). The noise current density i_n is essentially composed of shot noise and thermally induced gate-current noise. These noisy currents can normally be neglected for transistors using oxide thickness greater than 20 nm operated at low frequencies [20], [24]. Next, the gains of the noise sources are derived from the circuit model (Fig. 5), and the output noise power spectral density for the Miller integrator can be obtained

$$\begin{aligned} e_{no}^2 &= (e_{n1}^2 + e_{n2}^2 + e_{tR2}^2) \left| 1 + \frac{1}{s\tau} \right|^2 + e_{tR1}^2 \left| \frac{1}{s\tau} \right|^2 \\ &\quad + i_{n1}^2 \left| \frac{1}{sC_I} \right|^2 + i_{n2}^2 R_e^2 \\ &\approx e_n^2 + e_{tR2}^2 + e_{tR1}^2 \left(\frac{1}{2\pi f\tau} \right)^2 + i_{n1}^2 \left(\frac{1}{2\pi fC_I} \right)^2 + i_{n2}^2 R_e^2 \end{aligned} \quad (9)$$

where τ , the integrator time constant, is much greater than 1. Integrating e_{no}^2 over the system bandwidth, from f_{hp} to f_{lp} , gives the noise output power of the Miller integrator (v_{no}^2)

$$\begin{aligned} v_{no}^2 &= \int_{f_{hp}}^{f_{lp}} e_{no}^2 df \\ &= (e_n^2 + e_{tR2}^2 + i_{n2}^2 R_e^2) (f_{lp} - f_{hp}) \\ &\quad + \left(\frac{e_{tR1}^2}{\tau^2} + \frac{i_{n1}^2}{C_I^2} \right) \frac{f_{lp} - f_{hp}}{f_{lp} f_{hp}}. \end{aligned} \quad (10)$$

Equation (10) shows that both the MOS-bipolar resistor thermal noise (e_{tR1}), and the OTA input current noise (i_{n1}) undergo attenuation in the integrator. On the other hand, since R_e typically ranges from a hundred kilohms to few megaohms, and i_{n2} is typically equal to a few femtoamperes [27], the voltage noise source resulting from $i_{n2} R_e$ is disregarded, and e_n therefore dominates. As a result, the amplifier output noise power can be approximated by $e_n^2 (f_{lp} - f_{hp})$. Furthermore, having the input transistors drain current (I_{D1}) twice in $A2$ than in $A1$ results in a smaller input referred noise for $A2$ and makes the bioamplifier input-referred noise to be dictated by $A1$ input-referred noise ($v_{ni(A1)}^2$) only.

D. Noise Efficiency Factor

The noise efficiency factor (NEF) [5], [12], [15], can be used to quantify the noise versus power consumption tradeoff in this design. Following the above assertions, the bioamplifier NEF is given by

$$\text{NEF} = v_{ni(A1),\text{rms}} \sqrt{\frac{2I_{\text{Total}}}{4\pi \cdot U_T kT \cdot BW}} \quad (11)$$

where I_{Total} is the total supply current of the bioamplifier, and BW is the bandwidth. Denoting OTAs supply currents $I_{SS(A1)}$ and $I_{SS(A2)}$, we replace $I_{\text{Total}} = I_{SS(A1)} + I_{SS(A2)}$ and $I_{SS(A2)}/I_{SS(A1)} = \alpha$ in (11). Then, we can show that adding the *low-frequency suppression* circuit increases the NEF by $(1 + \alpha)^{1/2}$ compared to having $A1$ alone. In this design, α is chosen equal to 1.33 (2.66 μA /2.0 μA), which

results in an increase of 52% of the NEF. Nevertheless, *active low-frequency suppression* does not significantly increase the total input-referred noise, as shown in the previous section. Then, assuming that input transistors $M_{1,2}$ operate in weak inversion, $IC_1 \ll IC_3$, $IC_1 \ll IC_9$, $I_{\text{Total}} = 4(1 + \alpha)I_{D1}$, $\Delta f_{(A1)} = \pi/2 \cdot BW$, and $\Gamma = 1/2$, replacing (8) in (11), gives

$$\text{NEF} = \sqrt{\frac{4n^2(1 + \alpha)}{\varphi(IC_1)}} \approx 2n\sqrt{1 + \alpha}. \quad (12)$$

Equation (12) gives a theoretical limit for the NEF of the bioamplifier. With an average value of 1.5 for n in bulk CMOS processes [28], and $\alpha = 1.33$, a theoretical NEF of 4.6 is found for this design according to (12).

E. Stability

OTA $A1$ features several poles, with a dominant one located at $\omega_{p1} = g_o/C_L$. Similarly, this OTA has a pair of poles at $\omega_{p3,4} = g_{m3,4}/C_{3,4}$, and a pole-zero doublet at $\omega_{p10} = g_{m10}/C_{10}$ and $\omega_z = 2\omega_{p10}$, respectively. $C_{3,4}$ and C_{10} are the total capacitances seen at the gates of transistors $M_{3,4}$ and M_{10} . The proposed bioamplifier configuration (Fig. 1) presents a phase shift of -180° between nodes v_{in} and v_{out} . Also, poles and zeros of $A1$ introduce a negative phase shift, noted φ_{A1} . In addition, the Miller integrator (considered ideal) generates a positive phase shift of 90° in the loop gain (Fig. 2). Thus, the total phase shift introduced in the loop gain is $\varphi_{LG}(j\omega) = -180^\circ + \varphi_{A1} + 90^\circ$. At unity-gain frequency (ω_t), the phase margin (φ_M) of the whole bioamplifier is therefore

$$\varphi_M = \varphi_{LG}(j\omega_t) - 180^\circ = -\varphi_{A1} - 90^\circ \quad (13)$$

where $\varphi_{A1} < 0$. Hence, at ω_t , a phase shift of $\varphi_{A1} = -180^\circ$ in $A1$ yields a large φ_M of 90° , whereas a phase shift of $\varphi_{A1} = -90^\circ$ yields a null φ_M .

F. Low-Frequency Gain and Bandwidth Accuracy

As shown in (1), the gain and highpass cutoff frequency of the bioamplifier are defined by the open-loop gain A_{01} , which depends on the OTA transconductance parameter (g_m) and its output conductance (g_o). However, unlike g_m , g_{ds} may be difficult to predict because of its sensitivity to second order effects, including short-channel effects [22], [28], [29]. In a first order approximation, g_{ds} is proportional to the drain current over the early voltage (V_A)

$$g_{ds} = \frac{I_D}{V_A + V_{DS}} \approx \frac{I_D}{V_A}. \quad (14)$$

The early voltage roughly scales up with the transistor channel length (L) [21], at approximately 5 V/ μm for 0.18- μm bulk CMOS process. Drain-induced barrier lowering causes V_A to decrease at low IC, whereas the substrate current induced body effect, due to hot carriers and impact ionization, decreases V_A at high IC. On the other hand, V_A remains linearly related to L for long-channel devices presenting high inversion coefficients, bounded to 100 [28]. Consequently,

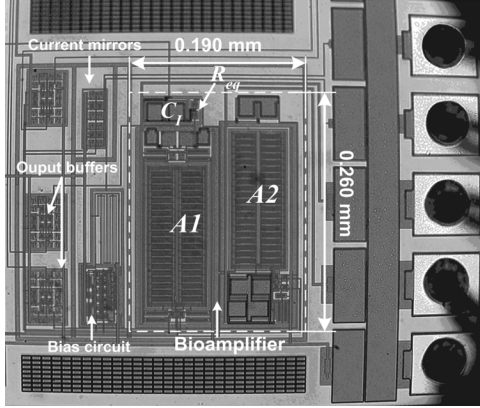


Fig. 6. Microphotograph of the fabricated chip. It includes one bioamplifier, a bias circuit with cascode current mirrors, unity-gain voltage buffers for driving loads, and on-chip current mirrors for testing.

long-channel transistors, featuring high inversion coefficients ($10 < IC < 100$), are needed for the output stages of A1 and A2. This design uses $L_{3,4,7,8,9,10} = 20 \mu\text{m}$, with inversion coefficients well above 10.

In addition, operating input transistors M_1, M_2 in weak inversion, and output transistors M_8, M_{10} in strong inversion minimizes dependency of A_{01} on drain currents, and therefore, on bias conditions. Replacing (14) in the low-frequency gain expression gives

$$A_{01} = g_m \cdot g_o^{-1} = \frac{I_{D1}}{nU_T} \cdot \frac{1}{I_{D8}} \left[\frac{V_{A8} + V_{A10}}{V_{A10}V_{A8}} \right]. \quad (15)$$

Then, equal drain currents $I_{D1} = I_{D8} = I_{D10}$ cancel out, and assuming V_A constant in a first order approximation, A_{01} becomes equal to $1/(nU_T) \cdot V_{A8}/V_{A10}$, which is independent of drain currents. In fact, since g_o and g_m are both linearly related to drain currents I_{D1} and $I_{D8}(I_{D10})$, any variation in I_{bias} attenuates and have little impact on the gain.

Following this, we emphasize that although simulated g_o may differ from measured values, satisfactory channel-to-channel gain matching can be obtained in a multichannel device by applying the design insights presented above. *Less gain variation might arise with amplifier topologies that use resistors or capacitors ratios in the feedback path, but these may require more circuit area and power as it will be shown next.*

IV. EXPERIMENTAL RESULTS

The bioamplifier was implemented and fabricated in a TSMC 0.18- μm one-metal one-poly CMOS process. It was designed for a gain of 400 and a bandwidth of 9.2 kHz, with a bias current of $1 \mu\text{A}$ for A1, and $2 \mu\text{A}$ for A2. The fabricated bioamplifier fits in $0.260 \text{ mm} \times 0.190 \text{ mm}$ chip area, which represents 31% of the targeted area per channel of 0.160 mm^2 . To gain a meaningful insight into the behaviour of the bioamplifier in an implantable sensor, a fully integrated bias circuit was used. It consists of a poly-resistor that sets a low current in a wide-swing current mirror [20]. A die photograph of the circuits is shown in Fig. 6.

TABLE II
CHIP-TO-CHIP MEASUREMENTS FOR THE FABRICATED BIOAMPLIFIERS

Parameters	Batch 1 (5 samples)		Batch 2 (5 samples)	
	Average	σ	Average	σ
Midband gain (dB)	50.1	± 0.60	48.8	± 0.30
High-pass cutoff freq. (Hz)	103.4	± 7.7	92.3	± 9.3
Low-pass cutoff freq. (kHz)	9.1	± 0.31	9.0	± 0.39
A1 bias current (A)	0.96	± 0.03	0.92	± 0.04
A2 bias current (A)	1.94	± 0.06	1.88	± 0.07
Total power cons (μW)	8.5	± 0.20	8.3	± 0.25

σ = standard deviation. The supply voltage is 1.8 V.

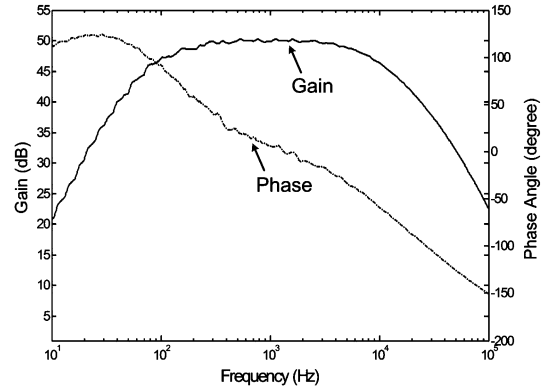


Fig. 7. Measured phase and frequency responses for the fabricated bioamplifier. The midband gain is 50.25 dB, and the highpass and lowpass cutoff frequencies are at 102 Hz and 9.1 kHz.

A. Integrated Bioamplifier Measured Performance

For assessing the chip-to-chip constancy of the presented bioamplifier design, testbench measurements were performed on two batches of chips, fabricated separately from two different wafers. Each batch included five packaged chips. Averaged measurements with standard deviations (σ) are reported in Table II. Fig. 7 presents the measured frequency and phase responses for one occurrence of the bioamplifier. The midband gain is 50.25 dB and cutoff frequencies f_{hp} and f_{lp} are located at 102 Hz and 9.1 kHz respectively, which yields a bandwidth of 9 kHz.

The measured input-referred noise voltage spectral density of the bioamplifier is shown in Fig. 8. The corner frequency is typically located around a few hundred hertz and the thermal noise level equals approximately $55 \text{ nV} \cdot \text{Hz}^{-1/2}$. Integrating under this curve gives a noise voltage of $5.5 \mu\text{Vrms}$. Fig. 9 shows a plot of the input-referred noise voltage for comparison. The noise voltage for this curve is $5.6 \mu\text{Vrms}$, which is consistent with measured spectral density shown on Fig. 8.

Average CMRR and power supply rejection ratio (PSRR) of 52 and 51 dB was respectively measured for all amplifiers. Terminals v_{in} and v_{ref} was tied to the same input signal for measuring the CMRR and an ac coupled input signal was injected at supply terminals VDD/VSS with v_{in} and v_{ref} grounded for measuring the PSRR. Total harmonic distortion (THD) remains under 1% for a 1-kHz input sine wave of amplitude smaller than 2.4 mV peak-to-peak. Hence, the bioamplifier exhibits a dynamic range of about 45 dB. Fig. 10 shows a plot of the total supply current versus the input-referred noise voltage for this

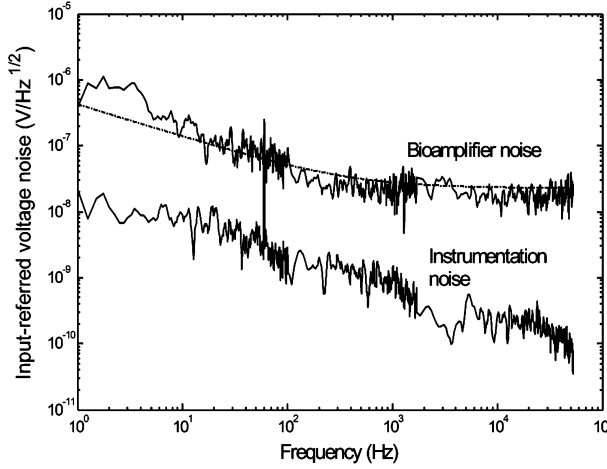


Fig. 8. Simulated (dashed) and measured (solid) input-referred noise voltage spectral densities. The corner frequency is around 300 Hz and the thermal noise is of $55 \text{ nV} \times \text{Hz}^{-1/2}$ at 100 Hz.

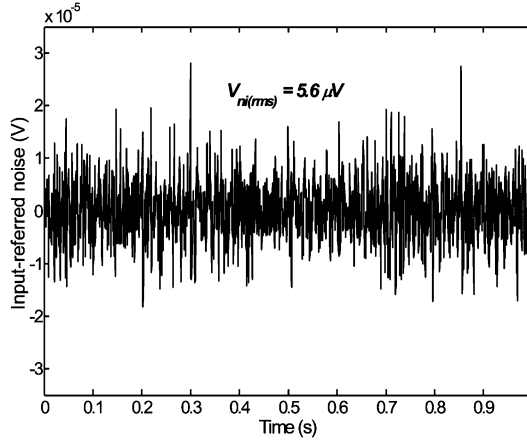


Fig. 9. Measured input-referred noise voltage of the fabricated bioamplifier (Output noise is divided by the gain with input terminals shorted to ground).

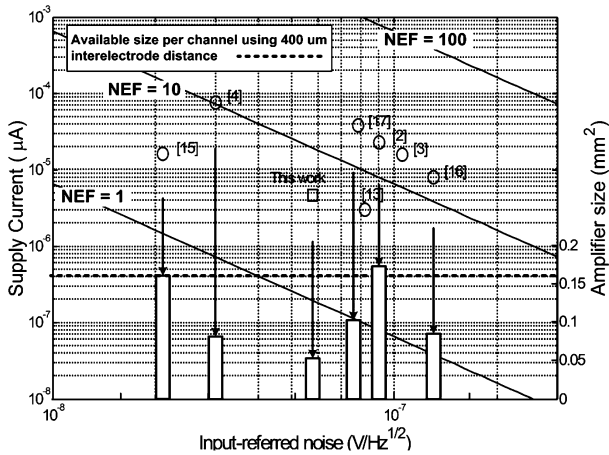


Fig. 10. Supply current versus noise for the fabricated bioamplifier (square) and for reported amplifiers (circles). Lines indicate constant NEF contours. The circuit size, illustrated with histograms, is also reported on the right y axis for comparison.

bioamplifier and for relevant previously reported bioamplifiers, also designed for AP recording. Histogram bars indicate the area

TABLE III
SIMULATED AND MEASURED PERFORMANCES OF THE
INTEGRATED BIOAMPLIFIER

Parameters	Simulation	Measured (Average $\pm \sigma$)
Midband gain	52 dB	49.52 ± 0.79 dB
High-pass cutoff frequency	105 Hz	$98.4 \text{ Hz} \pm 12 \text{ Hz}$
Bandwidth	9.2 kHz	$9.1 \text{ kHz} \pm 400 \text{ Hz}$
Input-referred noise	$5.4 \mu\text{V}_{\text{rms}}$	$5.6 \mu\text{V}_{\text{rms}}$
Noise efficiency factor	4.6	4.9
DC offset rejection	--	$\pm 450 \text{ mV}$
THD (2.4 mVpp at 1 kHz)	--	$< 1\%$
Dynamic range (THD $< 1\%$)	--	45 dB
CMRR (10 Hz – 9 kHz)	$> 45 \text{ dB}$	$52.68 \pm 1.05 \text{ dB}$
PSRR (10 Hz – 9 kHz)	$> 45 \text{ dB}$	$51.93 \pm 1.44 \text{ dB}$
Supply voltage	1.8 V	1.8 V
Supply current	$4.80 \mu\text{A}$	$4.67 \mu\text{A} \pm 0.15 \mu\text{A}$
Area (in 0.18- μm CMOS)	--	0.050 mm^2

occupied by each bioamplifier with respect to the y axis at right. As shown, a measured NEF of 4.9 is achieved for our bioamplifier with a silicon area of 0.050 mm^2 . Thus, our design achieves both the lowest reported size and one of the best NEFs. Table III summarizes simulated and measured performances for the integrated bioamplifier. Differences between the simulated and measured midband gain and cutoff frequencies are likely due to lower bias currents achieved by the integrated bias circuits compared to simulated values (Table II). This also resulted in lower power consumption measured for both OTAs.

B. In Vivo Neural Recordings

The proposed bioamplifier was used for neural recordings in an *in vivo* experiment with rats. A stainless steel-tipped micro-electrode with a shank diameter of $75 \mu\text{m}$ (FHC Inc., Bowdoin, ME) was used to record extracellular APs from the hippocampus of an anesthetised rat. Spontaneous activity was retrieved by the electrode with the bioamplifier. The electrode was connected to the input terminal v_{in} and a driving screw, used for providing a reference voltage, was connected to the terminal v_{ref} . The screw was inserted into the skull, a few millimetres from the measurement site. Shielded wires was used for both connections.

The bioamplifier output was connected to a commercial rack mounted amplifier (A-M systems, Inc., Carlsborg, WA) for further amplification and filtering. An additional gain of 100 was provided and signals were bandpass filtered between 300 Hz and 10 kHz. The resulting signals were sampled and digitized at a rate of 25 kHz, using commercial data acquisition hardware and operating software (DataWave Technologies, USA). Fig. 11(a) shows a neural signal recorded extracellularly from the rat hippocampus with the presented bioamplifier. A closer view on a detected AP is shown in inset. Fig. 11(b) shows several recorded APs isolated from the neural signal and aligned on top of one another.

V. CONCLUSION

A novel integrated bioamplifier featuring *active low-frequency suppression* was presented. The topology uses a Miller integrator connected within the feedback path of a low-noise OTA for closed-loop control of the dc level. The working principle of the proposed amplifier was demonstrated through the design of a bioamplifier for neural recording of APs.

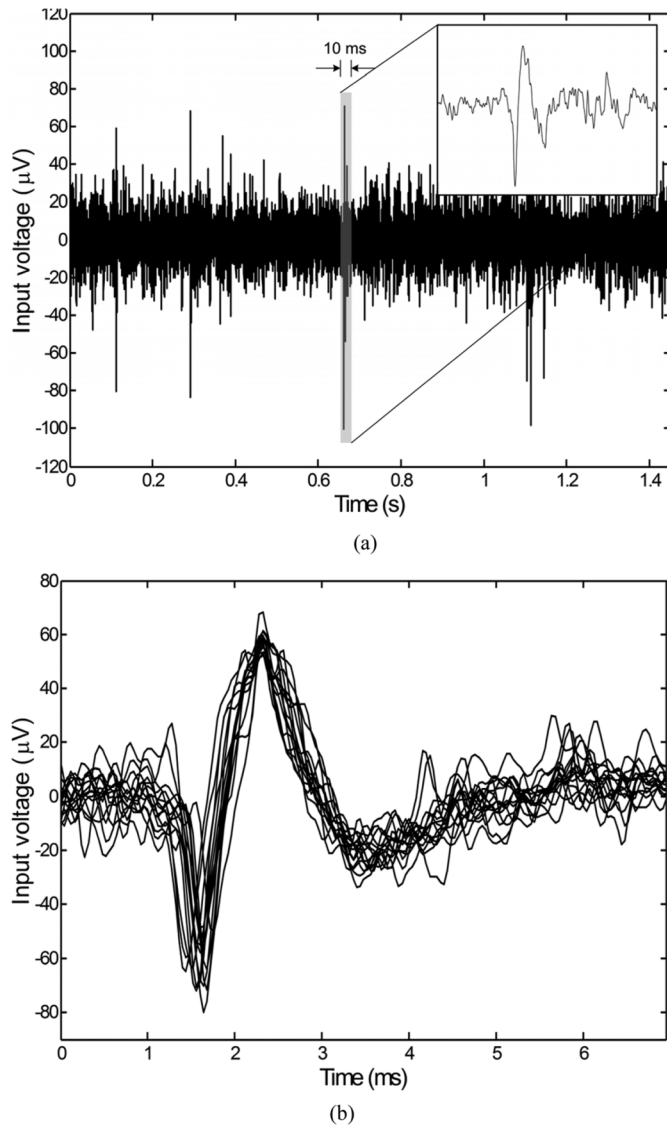


Fig. 11. Extracellular recording from the hippocampus of a rat. (a) Sample of a waveform recorded with the presented bioamplifier. A closer view on an AP is shown in inset. (b) Fifteen similar APs isolated from the recorded waveform and aligned on top of one another. (Waveforms are referred to the amplifier input).

The circuit uses two micropower OTAs optimized through a transconductance efficiency-based design methodology. A MOS-bipolar device was used for implementing a high-value equivalent resistor coupled with a small integrated capacitor of 1 pF to set the integrator time constant, yielding a highpass cutoff frequency of 100 Hz. The bioamplifier was fabricated in a TSMC 0.18- μm CMOS process and was successfully tested. Typically, it features a power consumption of 8.6 μW , an input-referred noise of 5.6 μV_{rms} , a midband gain of 50 dB, and a bandwidth of 9 kHz. Moreover, the bioamplifier was validated successfully by *in vivo* measurements in which extracellular neural activity from the rat hippocampus was recorded.

The bioamplifier presented in this paper addresses power and size-budget requirements for increasing the channel count in multichannel implantable recording devices. Moreover, it leaves room for additional on-chip modules, such as data

converters, biopotential detectors, embedded controllers, and integrated telemetry systems, within typical channel integration size. Thus, the proposed design is fully compliant for massive integration in high channel-count neural sensors.

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Benoît Gosselin (S'02) received the B.Sc. degree in electrical engineering from École Polytechnique de Montréal, Montréal, QC, Canada, in December 2001, where he is currently working toward the Ph.D. degree under the direction of Prof. M. Sawan.

He was a research assistant at the Defence Research Establishment in Ottawa, ON, Canada, in 2001, and he was involved in the design and test of a neuroelectric sensor for Victhom Human Bionics Inc. (Québec, QC, Canada) during his graduate studies. His thesis topic involves the design and implementation of low-power neural integrated interfaces where he is interested in low-noise amplifiers, low-power circuit techniques and biotelemetry.



Mohamad Sawan (S'88–M'89–SM'96–F'04) received the B.Sc. degree from Université Laval, Québec, QC, Canada, in 1984, and the M.Sc. and Ph.D. degrees from Université de Sherbrooke, Sherbrooke, QC, Canada, in 1986 and 1990, respectively, all in electrical engineering.

He conducted postdoctoral training at McGill University, Montreal, QC, Canada, in 1991. He joined Ecole Polytechnique de Montréal in 1991, where he is currently a Professor in microelectronics.

His scientific interests are the design and test of mixed-signal (analog, digital, and RF) circuits and systems, digital and analog signal processing, and modeling, design, integration, assembly, and validation of advanced wirelessly powered and controlled monitoring and measurement techniques. These topics are oriented toward biomedical implantable devices and telecommunications applications. He holds the Canadian Research Chair in Smart Medical Devices. He is leading the Microsystems Strategic Alliance of Québec (ReSMiQ). He has published more than 350 papers in peer-reviewed journals and conference proceedings. He holds six patents.

Dr. Sawan is Editor of *Mixed-Signal Letters* and Associate Editor of the IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS (TBCAS). He is a Fellow of the Canadian Academy of Engineering. He is Founder of the Eastern Canadian IEEE Solid-State Circuits Society Chapter, the International IEEE-NEWCAS conference, cofounder of the International FES Society, and Founder of the Polystim Neurotechnologies Laboratory, Ecole Polytechnique de Montréal. He was a Distinguished Lecturer for the IEEE Circuits and Systems Society between 2003 and 2006. He received the Barbara Turnbull 2003 Award for spinal cord research, the Medal of Merit from the President of Lebanon, the Bombardier Medal of Merit from the French Canadian Association for the Advancement of Sciences and the American University of Science and Technology Medal of Merit.



C. Andrew Chapman received the B.Sc. degree in psychology from the University of Toronto, Toronto, ON, Canada, after having worked in the laboratory of J. Yeomans. His graduate work with R. Racine at McMaster University (1995), Hamilton, ON, Canada, focused on factors that regulate changes in the strength of synapses in the entorhinal cortex and in the sensorimotor cortex of awake animals.

He then moved to the Université de Montréal, Montreal, QC, Canada, to conduct postdoctoral work with J.-C. Lacaille, and used intracellular recordings from individual neurons in acute brain slices to investigate synaptic alterations and rhythmic activity in hippocampal inhibitory interneurons. He joined the Department of Psychology at Concordia University, Montreal, QC, Canada, in 1999, and has been an Associate Professor since 2004.